NAO VI Guide

Explanation of Flowchart

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For:

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Introduction

This document provides an explanation of the flowchart illustrated in the accompanying *Flowchart* PDF. The purpose of this document is to describe the sequence of actions, decision points, and logical flow used in the NAO robot's interactive behavior system. Each section of the chart—covering startup, autonomous routines, tactile and voice command inputs, and response actions—is detailed to ensure a clear understanding of how the system transitions between different states during operation. This explanation aims to serve as a reference for development, debugging, and demonstration within the NEST Lab at the University of British Columbia.

Flowchart

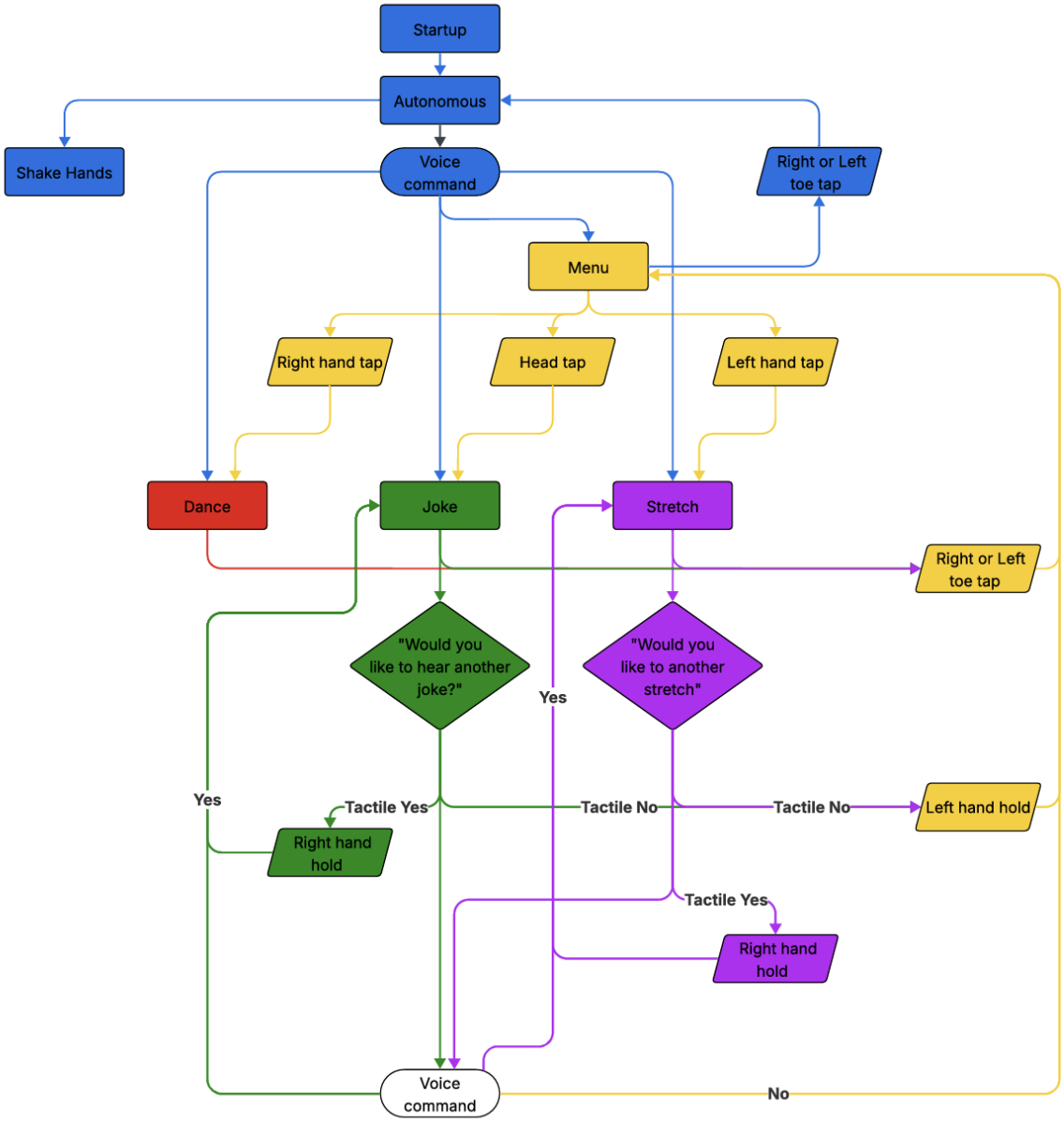

Figure 1 Flowchart for NAO behaviors

Instructions

Blue Section – Initialization and Core Control

🟦 The process begins at the **Startup** stage, where the NAO robot powers on and initializes all systems. It then enters the **Autonomous** mode, allowing it to independently recognize and respond to both tactile and voice inputs. In this mode, the robot can directly take **voice commands** to perform actions such as 🟥 **Dance**, 🟩 **Joke**, or 🟪 **Stretch** without requiring manual input.
If the user says the keyword **"Menu,"** the robot transitions into a selection interface where tactile gestures are used to choose activities: a **Right Hand tap** activates **Dance**, a **Head tap** triggers **Joke**, and a **Left Hand tap** starts the **Stretch** routine. This design enables both hands-free operation through speech and manual selection via tactile interaction, providing flexible user control.

Note: The system does *not* return directly to Voice Command after routines. All routines exit to the Menu first, and returning to Voice Command requires additional toe presses.

Yellow Section – Menu and Gesture Inputs

🟨 The Menu section is the central hub for physical interactions. It can be reached from a voice-triggered "Menu" command or from a toe-tap exit from any routine. Within this section, the robot responds to three tactile triggers: a **Right Hand tap** starts the **Dance** routine, a **Head tap** begins the **Joke** routine, and a **Left Hand tap** initiates the **Stretch** routine.
From this Menu state, the user may reach Voice Command only after additional toe presses, following the new multi-press exit rules.

Red Section – Dance Routine

🟥 The Dance routine is activated by a voice command or a **Right Hand tap**. During the dance, the user may use **toe taps** to interrupt the routine. When the dance finishes—or when interrupted—the robot returns to the **Menu**. From the Menu, additional toe presses are required to reach Voice Command.

Toe tap logic:

- **1 toe press → Menu**
- **2 toe presses total → Voice Command**

Green Section – Joke Routine

🟩 When the user triggers the **Joke** routine (by **voice** or **Head tap**), NAO tells a joke and then asks, *"Would you like to hear another joke?"* If the tactile response is **Yes** (through **voice** or a **Right Hand hold**), the robot loops back to the joke stage to tell another one. If the tactile response is **No** (**Voice** or **Left Hand Hold**), control returns to the **Menu** stage for a new instruction. This green segment captures the humor and dialogue interaction pathway. A **Right or Left toe tap** at this stage can also redirect the system to the **Menu**. Additional toe presses are required to reach **Voice Command**.

Purple Section – Stretch Routine

🟪 The **Stretch** routine begins either from a **Left Hand tap** or from **Voice**. After completing one stretch cycle, NAO asks, *"Would you like to do another stretch?"* A **Yes** response (**Voice** or **Right Hand hold**) restarts the stretching motion, while a **No** (**Voice** or **Left Hand hold**) response transitions control back to **Menu**. Additionally, a **Right or Left toe tap** at this stage can also redirect the system to the **Voice Command**, allowing users to easily switch tasks.